BigBasket Product Pricing & Category Analysis

Adata-driven explorationof **pricing**, **discounts**, and**customerratings** across India's largestonline grocery platform— **23,000+**product listings analyzed.

AmitiAneesh

Tableau dashboard, KPI computation

Lakshith

Exploratory & Statistical Analysis

Chanakya Sinde

Data Cleaning & ETL Pipeline

Sai Praneeth Sharma

Business Insights & Final Report

DOMAIN: E-COMMERCE / RETAIL

PYTHON · TABLEAU · GITHUB

REPO: CAPSTONE2

The Problem We Set Out to Solve

CORE QUESTION

Core Question

What are the key drivers of product pricing, discount patterns, and customer ratings across categories and brands on BigBasket?

BUSINESS GOAL

Business Goal

- **Optimize** pricing and discount strategies
- **Identify** high-performing brands and categories
- **Understand** what drives higher customer ratings
- **Improve** product assortment decisions

Project Overview

Thisproject analyzesthe **BigBasketEntireProduct List Dataset** — a real-world catalogfrom India'slargestonlinegrocery supermarket, sourced from Kaggle.

Pricing Trends

Sale vs. market price across **categories** and **sub-categories**

Discount Patterns

Discount depth distribution and segment-level breakdowns

Customer Ratings

Relationship between **pricing**, **brand**, and consumer satisfaction

Brand Performance

Top brands by **discount rates**, **ratings**, and catalog presence

Dataset at a Glance

23K+

Product **Listings**

9

Raw **Columns**

5

Engineered **Features**

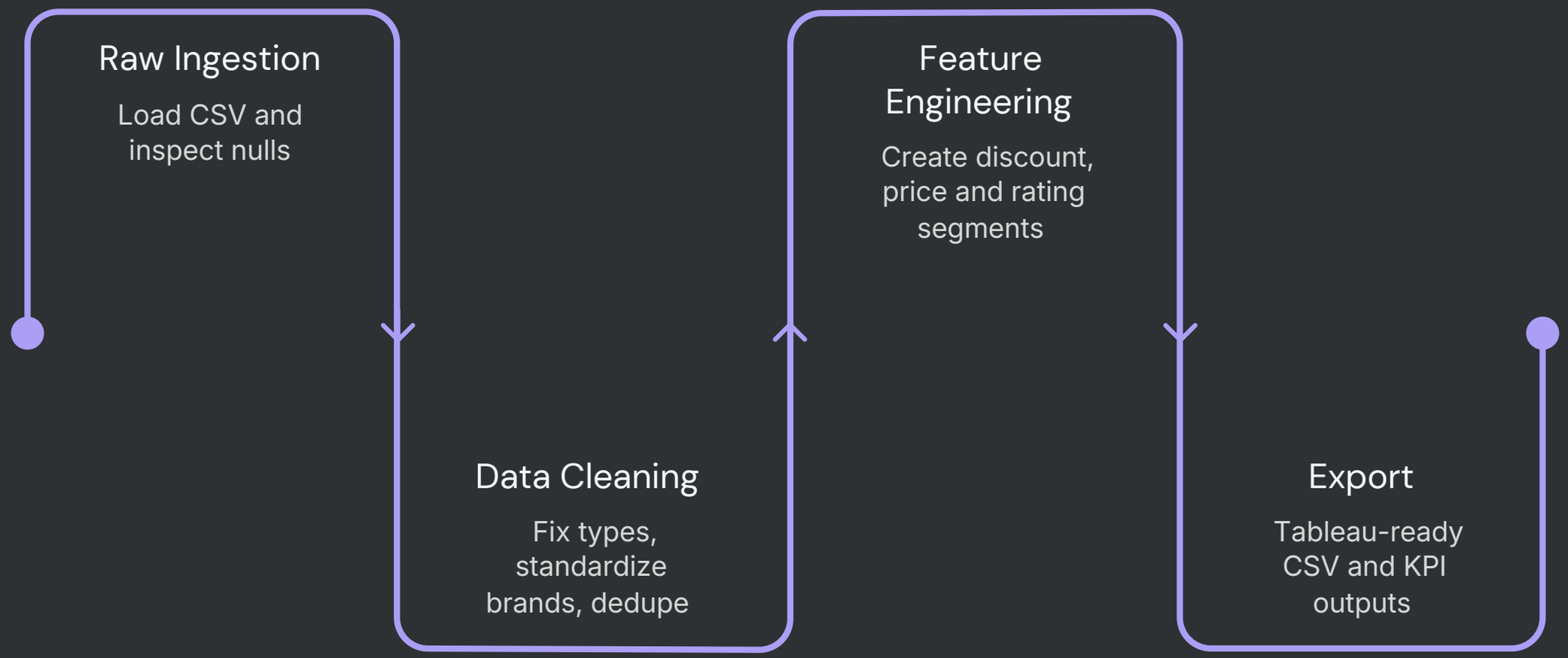
Key Raw Columns

- **sale_price / market_price** — pricing in ₹
- **category / sub_category** — catalog hierarchy
- **brand** — manufacturer/seller name
- **rating** — consumer score out of 5
- **type** — product type classification

Engineered Features

- **discount_amount**, **discount_pct**, **price_segment**
- **is_discounted**, **rating_segment**

Data Preprocessing & ETL Pipeline



Handled in `02_cleaning.ipynb` — price fields converted from ₹-formatted strings to floats, anomalous records flagged, brand names standardized to title case, and five new analytical features engineered.

Analysis Performed

1

Pricing & Discount Analysis

Avgsalevs. market pricebycategory; discount % distribution; **Budget / Mid-range / Premium** segmentation

2

Category & Brand Performance

Top categoriesby productcountandavg rating; brand-level discount and rating comparison

3

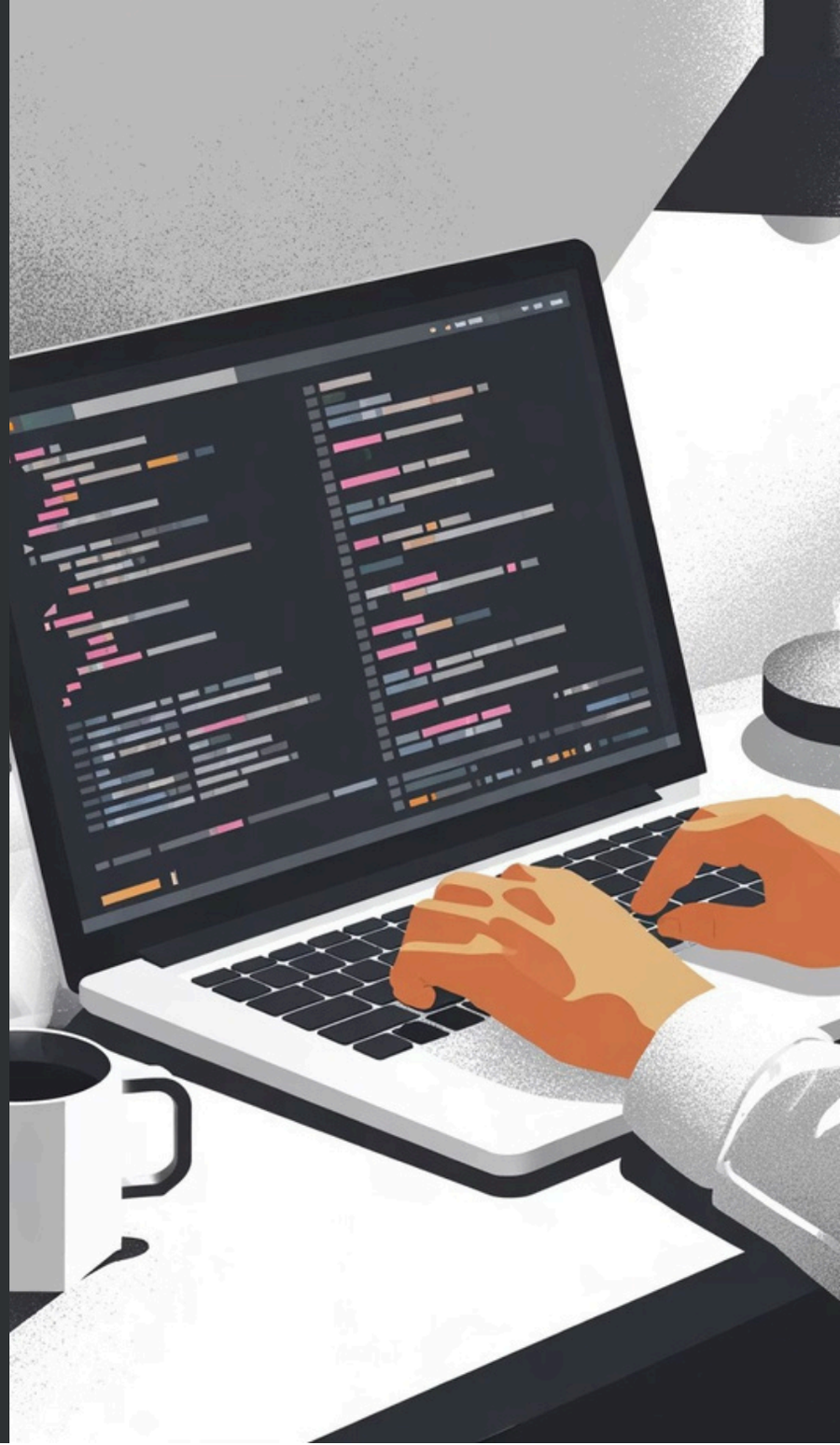
Rating Analysis

Ratingdistribution; correlation with discount %; category-wise avg rating; **high vs. low-rated profiling**

4

Statistical Analysis

Correlation matrix, hypothesis testing, regression on rating predictors, **ANOVA across categories**



KPIs Tracked

Avg Discount %
Mean discount across all products

Avg Sale Price
Mean sale price across the full catalog (₹)

Avg Rating
Mean consumer rating across all listings

Top Brand
Best-rated brand across the catalog

Price Segment Split
Share of Budget / Mid-range / Premium products

% Discounted
Share of products priced below market rate

Key Business Insights

Discount ≠ Satisfaction

Products with **moderate discounts (10–30%)** rate better than heavily discounted ones — deep cuts may signal poor quality.

Budget Segment Dominates

Budget-segment products hold the **highest volume** but carry lower average ratings — a quality-gap oppor tunit y.

Price ≠ Quality

Premium-priced products do **not consistently earn higher ratings** — price alone is not a reliable quality signal.

Assortment Gaps

Several sub-categories show **very low average ratings**, signaling underperforming assortments ripe for curation.

Brand Concentration

A **small number of top brands** dominate across multiple categories, creating a long-tail distribution in the catalog.

Missing Rating Patterns

Missing ratings concentrate in **newer or less-purchased product types** — an onboarding and engagement signal.

Business Recommendations

Revisit Deep Discount Strategy
Low-rated, heavily discounted products may be eroding brand trust — consider pulling or repositioning them.

Improve Product Descriptions
Sub-categories with low ratings often have poor descriptions — better content sets accurate expectations.

Promote High-Rating Mid-Range Brands
Feature these brands on the homepage and in targeted campaigns for maximum satisfaction-to-cost ratio.

Curate Underperforming Sub-Categories
Reduce low-rated SKUs and replace with better-reviewed alternatives to lift category averages.

Segment-Targeted Promotions
Use **Budget vs. Premium** price segmentation to design personalized promotions that resonate with each audience.

Team Contributions & Tech Stack

Contribution Matrix

Member	Role & Deliverables
Amiti Aneesh	**Tableau dashboard, KPI computation**
Lakshith	**EDA, trend analysis, statistical tests**
Chanakya Shinde	**Data ingestion, ETL pipeline, feature engineering**
Sai Praneeth Sharma	**Business insights, recommendations, final report**

Tech Stack & Repository

Python

ETL, EDA, Statistics

Tableau

Interactive Dashboard

GitHub

Repo: capstone2

Repository structure includes `data/`, `notebooks/`, `tableau/`, `docs/`, and `reports/final_report.pdf`.